1997 S.C.E. HIGHER I (ANSWERS)

1. $2x + y = -3$

2. Verify that $\vec{AB} = \begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix}$ $\vec{BC} = \begin{pmatrix} 2 \\ -8 \\ 4 \end{pmatrix}$

$\therefore \vec{AB} = \frac{1}{2}\vec{BC}$

$\therefore$ AB is parallel to BC

$\therefore$ A, B, C are collinear ($\because$ of the common point B)

3. $f[g(x)] = 2(\sin x + \cos x)$

$g[f(x)] = \sin 2x + \cos 2x$

4. (a) $\vec{PQ} = \begin{pmatrix} 8 \\ -4 \\ 1 \end{pmatrix}$ (b) $PQ = 9$ units

5. (a) A real root is $x = 2$ (b) Other roots come from $2x^2 + x + 4 = 0$

$\therefore \Delta = -31$

Since $\Delta < 0$, there are no other real roots

6. $x = -1$

7. [You will need to construct a Δ from $\tan x = \frac{4}{3}$].

8. $\frac{dy}{dx} = 4x + 1$; then replace $\frac{dy}{dx}$ by $4x + 1$ in $x\left(1 + \frac{dy}{dx}\right)$.

9. (a) Prove that $f(x) = 2(x+2)^2 - 11$ (b) MIN T.P. is $(-2, -11)$

10. $\frac{14}{3}$

11. $\sqrt{29}\sin(x - 68{\cdot}2^\circ)$

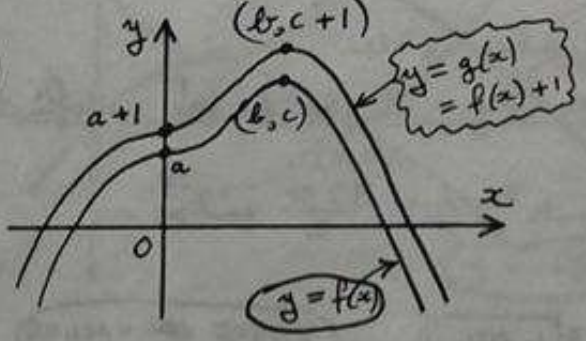
12. $(x-13)^2 + (y-4)^2 = 17$

13. $\underline{a} \cdot (\underline{b} + \underline{c}) = 0$; $\underline{a}$ is perpendicular to $\underline{b} + \underline{c}$.

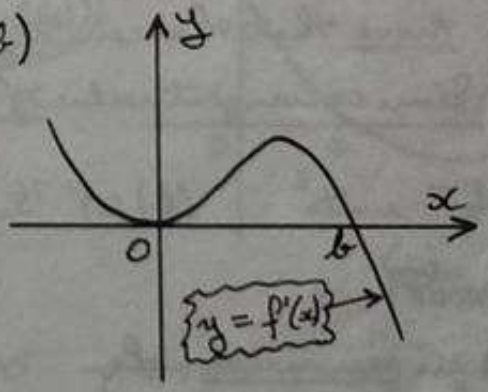
14. $c < 13$

15. $f(x) = \frac{1}{2}\sin 2x + \frac{3}{4}$

16. (a)



(b)



(c) g'(x) is the same graph as f'(x)

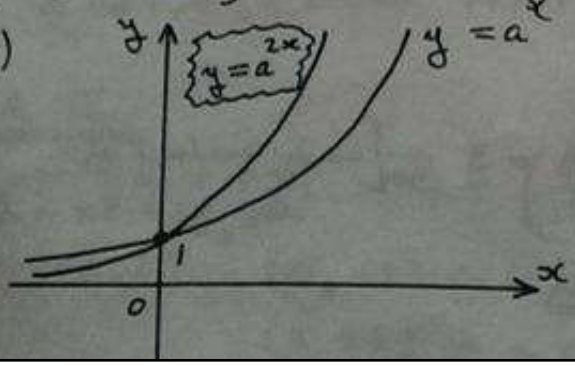
17. $A(-4\frac{1}{2}, 0)$ $B(14{\cdot}91, 8)$

18. (a) [Replace $\cos 2x$ by $1 - 2\sin^2 x$ and $\cos^2 x$ by $1 - \sin^2 x$]

(b) $x = 19{\cdot}45^\circ, 160{\cdot}55^\circ, 270^\circ$

19. (a) $t = a$; $u = 0$

(b)



(c) Point of intersection is $(1, a^2)$

20. $a = 51{\cdot}34^\circ$.

HIGHER II

(a) $A(1,3)$ $B(-3,-5)$ (b)(i) $C(-5,1)$ (ii) $x+2y=-3$

∴ (a) $R(7,-1,6)$ (b) $\left[\overrightarrow{SR}=\begin{pmatrix}9\\-3\\1\end{pmatrix}\ \overrightarrow{SP}=\begin{pmatrix}1\\1\\-3\end{pmatrix}\right]$ $P\hat{S}R = 84{\cdot}56°$

3. (a) $u_0(1^{ST}\text{JAN}) = 1000$ $u_1(1^{ST}\text{FEB}) = 1105$ $u_2(1^{ST}\text{MAR}) = 1210{\cdot}53$
$u_3(1^{ST}\text{APRIL}) = 1316{\cdot}58$ $u_4(1^{ST}\text{MAY}) = 1423{\cdot}16$ $u_5(1^{ST}\text{JUNE}) = 1530{\cdot}28$ $u_6(1^{ST}\text{JULY}) = 1637{\cdot}93$

∴ Amount on JUNE 30TH = £1537·93

(b) [Continue until $u_{10}(1^{ST}\text{NOV}) = 2073{\cdot}94$] ∴ First exceeds £2000 on 1ST NOVEMBER

(c) $u_{n+1} = 1{\cdot}005\,u_n + 100$ where u_n is the amount in account n months after 1ST JAN on 1ST of month (where $u_0 = 1000$).

4. (a) $P(1,0)$ $Q(2,0)$ (b) [TOTAL AREA $= 1\frac{11}{12} + \frac{7}{12}$] $= 2\frac{1}{2}$ units^2

5. (a) $A(0,5)$ $B(2,1)$ (b) [FIRST PROVE THAT CURVES MEET WHEN $x=0$ or 4]
∴ AREA ENCLOSED $= 21\frac{1}{3}$ units^2

(c) $m = 10$, $n = -1$

6. (a) $m_{TGT} = -\frac{1}{a^2}$ (b) — (c)(i) [PROVE THAT B IS $(0, \frac{2}{a})$ AND C IS $(2a, 0)$]
∴ AREA OF $\triangle OBC$ $= 2$ units^2
(ii) Area of $\triangle OBC$ is always 2 units^2 regardless of position of tangent at A

7. $\frac{5x+1}{(x-4)(x+3)} = \frac{3}{x-4} + \frac{2}{x+3}$

8. (a) $k = -0{\cdot}000122$ (b) 88·5%

9. (a) LIKELIEST : $\lambda = 3$ $\varrho = 1$ $\nu = -140°$ $u = 230°$
OTHERS : $\lambda = 3$ $\varrho = 1$ $\nu = 220°$ $u = 230°$
$\lambda = -3$ $\varrho = 1$ $\nu = 40°$ $u = 230°$
$\lambda = -3$ $\varrho = 1$ $\nu = -320°$ $u = 230°$

(b) $\alpha = -0{\cdot}928$
$t = 120{\cdot}53°$
[USE EQUATION WITH REPLACEMENTS FOR λ, ν AND ϱ]

10. (a)(i) [USE SIMILAR $\triangle$'s ADE AND ABC] (ii) [$V = (2x)(2a)(10 - \frac{5}{2}x)$ etc]
(b) [PROVE THAT GREATEST VOLUME OCCURS WHEN $x = \frac{8}{3}$ (not $x=0$ ∵ $x>0$)]
∴ DIMENSIONS ARE $\frac{16}{3}$ cm × $\frac{16}{3}$ cm × $\frac{10}{3}$ cm

11. (a)(i) — (ii) $OC = 1 + 2\cos\lambda$ [and finally use $d^2 = OB^2 + OC^2$]
(b)(i) $d^2 = 6 + 4\sqrt{2}\cos(\lambda - \frac{\pi}{4})$ (ii) MAX. VALUE OF d^2 IS $6+4\sqrt{2}$; OCCURS WHEN $\lambda = \frac{\pi}{4}$.
(c)(i) [REPLACE λ by $\frac{\pi}{4}$ in $OB = 1 + 2\sin\lambda$]; $BD = \frac{1}{2}(\sqrt{2}+2)$
(ii) $\sqrt{6+4\sqrt{2}} = 2 + \sqrt{2}$